Practice Test: Chapter 1 (Atoms & Elements)

Name: ______________________________ Date: ____________

No.	Question /Answer
1	How many protons, neutrons and electrons are in the ion $^{71}_{33}Ga^{3+}$?
2	An ion contains 11 protons and 13 neutrons in the nucleus, which is surrounded by 10 electrons. Give the formula of this ion.
3	A selenide ion, Se^{2-}, has a mass number of 78. How many protons, neutrons and electrons are in this ion?
4	Two isotopes of chlorine are $^{35}_{17}Cl$ and $^{37}_{17}Cl$. Which of the following statements about these two atoms are correct? **A** They have the same number of protons. **B** Both atoms have the same mass number. **C** The lighter isotope has 35 neutrons but the other has 37 neutrons. **D** Both isotopes will react with sodium to form ions with a -1 charge.
5	Give the electron configuration of the following species. **a** Ca **b** P^{3-} ion **c** aluminium ion **d** H^+ ion
6	A neutral atom has the electron configuration 2, 8, 3. In which period and group of the periodic table would this element be located?
7	Give the formulas of two negative ions with an electron configuration of 2, 8.
8	What forces are responsible for keeping the atom stable? Describe how these forces interact.
9	Where are valence electrons found?
10	For each of the following, state the main energy level of the valence shell and the number of valence electrons: **a** fluorine **b** calcium ion **c** aluminium **d** bromide ion
11	Fluorine is in the same group as chlorine. Explain why fluorine has a higher electronegativity than chlorine.

No.	Question /Answer
12	Explain why an oxygen atom is smaller than a sulfur atom.
13	Explain why a magnesium atom is larger than a magnesium ion
14	A silver atom is about nine times heavier than a carbon-12 atom. What would be the relative atomic mass of the silver atom?
15	The noble gas neon is well known for its emitted orange light seen in 'neon signs'. It occurs naturally as three isotopes with the following abundances: Ne-20 is 90.9%, Ne-21 is 0.3 %, and Ne-22 is 8.8%. Calculate the relative atomic mass of the element neon.
16	Naturally occurring copper consists of two isotopes with relative atomic masses of 62.93 and 64.93. If the percentage abundance of the lighter isotope is 69.15%, what is the relative atomic mass of naturally occurring copper?
17	Describe how the concentration of sodium ions is measured using atomic absorption spectroscopy. Use a schematic diagram to illustrate the process.
18	Describe how the relative abundance of isotopes of magnesium is determined using mass spectrometry. Use a flow diagram to illustrate how the different isotopes in a magnesium sample are separated and quantified.